Centroids of parabola-inscribed Poncelet polygons

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Abstract

The vertex, perimeter, and area centroids of the focus-circle Poncelet family trace coaxial parabolas for every polygon size. Their focal distances are, respectively, f/N, $f/2$, and $f/3$, where f is the outer parabola's focal distance. The proof extends the known ellipse centroid theorem by a controlled projective limit. In particular, it settles Conjecture 4 of Bellio–Garcia–Reznik and its pentagon special case.

Task 1: the original problem

Normalize the outer parabola, its focus, and the circular caustic as

$$\mathcal{P}:\ y^2+4fx=0, \qquad F=(-f,0), \qquad \mathcal{C}:\ (x+f)^2+y^2=r^2, \qquad 0<r<f.$$

We consider a closed Poncelet family with $N\geq 3$ distinct vertices: successive edges touch the inner circle. This is the circumscribed family studied in [1], including all its convex families in Table 2. Points at infinity are limiting configurations, rather than finite polygons at which a Euclidean centroid is defined. Write C_0, C_1, C_2 for the centroids of the vertices, the uniformly weighted edges, and the homogeneous polygonal lamina. For a star traversal of a nested pair the area centroid is understood with signed area and winding multiplicity, as in [2]; the same proof applies. The separate intersecting-conic configurations mentioned in the caption of Table 2 of [1] are not circumscribed polygons with an incircle.

Theorem 1. *There are real constants v_0, v_1, v_2, depending on the fixed family, such that the complete finite centroid loci are*

$$C_0:\ X=v_0-\frac{N}{4f}Y^2, \qquad C_1:\ X=v_1-\frac{1}{2f}Y^2, \qquad C_2:\ X=v_2-\frac{3}{4f}Y^2.$$

Furthermore $v_1=\frac{3}{2}v_2+\frac{1}{2}f$. Thus all three loci are nondegenerate parabolas with the same axis as $\mathcal{P}$, and their focal distances are $f/N, f/2, f/3$.

The input from earlier work is the following *ellipse* theorem: for nested ellipses admitting a Poncelet family, both C_0 and C_2 trace ellipses homothetic to the outer ellipse, or single points [2, Theorem 1]. The removable-singularity issue for the area centroid in that proof is addressed in [3, Proposition 2]. Neither the input theorem nor the tangential-polygon centroid identity below is claimed as new here. The source retains the general parabolic assertion as Conjecture 4 in both its current preprint and journal version.

The infinite-vertex limit

The real projective completion of $\mathcal{P}$ is a circle. The Poncelet step is a fixed-point-free, orientation-preserving circle map of order N on the chosen primitive family. One can see this by mapping the nested pair to nested ellipses. Consequently every orbit has N distinct vertices, including the orbit through the unique point I of $\mathcal{P}$ at infinity.

Let one vertex be

$$Z(t) = (-t^2/(4f), t), \qquad t \longrightarrow \pm\infty.$$

All other vertices tend to finite, distinct limits. The two edges incident with I are the horizontal tangents $y = r$ and $y = -r$ to $\mathcal{C}$. Hence the two neighboring vertices tend to

$$A_+ = (-r^2/(4f), r), \qquad A_- = (-r^2/(4f), -r).$$

Continuity here is projective continuity of the Poncelet map; the other orbit points are different from I and therefore remain bounded in affine coordinates. It follows immediately that

$$(C_0)_x = -\frac{t^2}{4fN} + O(1), \qquad (C_0)_y = \frac{t}{N} + O(1). \tag{1}$$

For the area centroid, split the polygon into the triangle formed by $Z(t)$ and its two neighbors and the remaining bounded polygon. This is also a valid identity for signed areas and first moments. The triangle's area has magnitude

$$\frac{r}{4f}t^2 + o(t^2).$$

The remaining polygon has bounded area and first moments. The triangle centroid is $Z(t)/3+O(1)$, and replacing its area by the total area changes the resulting centroid by $O(1)$. Therefore

$$(C_2)_x = -\frac{t^2}{12f} + O(1), \qquad (C_2)_y = \frac{t}{3} + O(1). \tag{2}$$

In particular, neither of the two centroid loci is contained in a line: a relation $aX + bY + c = 0$, first divided by t^2 and then by t, would force $a = b = c = 0$.

A projective limit that preserves the quadratic part

For $\varepsilon > 0$ set

$$T_\varepsilon(x, y) = \left(\frac{x}{1 - \varepsilon x}, \frac{y}{1 - \varepsilon x}\right).$$

The image of the projectively completed parabola is the ellipse

$$\Gamma_\varepsilon : \quad 4f\varepsilon X^2 + Y^2 + 4fX = 0. \tag{3}$$

Indeed the inverse substitution is $x = X/(1 + \varepsilon X)$, $y = Y/(1 + \varepsilon X)$. The map takes the closed parabolic region $y^2+4fx \le 0$, completed by its point at infinity, to the closed ellipse disk. The circle disk is strictly inside that region: on $\mathcal{P}$ the squared distance to F equals $(f + y^2/(4f))^2 \ge f^2 > r^2$. Thus $T_\varepsilon(\mathcal{C})$ is an ellipse strictly inside Γ_ε. The same projective transformation takes every polygon of our family to a Poncelet N-gon for this nested pair.

Fix $k \in \{0, 2\}$, and let $C_{k,\varepsilon}(P)$ be the centroid *computed after transforming all the vertices* of P. We do not assert projective covariance of centroids. For each finite polygon P, the vertex and area formulas give

$$C_{k,\varepsilon}(P) \longrightarrow C_k(P). \tag{4}$$

The denominator for C_2 is nonzero. In the convex case this is immediate; in a nested star family orient the edges so that the caustic is on their left, and the fan triangles from an interior point have positive signed area.

Choose three finite polygons for which the C_k values are noncollinear, using (1) or (2). Their transformed centroid values remain noncollinear for all sufficiently small $\varepsilon > 0$. The ellipse theorem then excludes the single-point alternative and gives one equation, valid for every polygon P,

$$4f\varepsilon X^2 + Y^2 + a_\varepsilon X + b_\varepsilon Y + c_\varepsilon = 0, \qquad (X, Y) = C_{k,\varepsilon}(P). \tag{5}$$

This normalization of the quadratic terms follows precisely from homothety to (3).

Substitute the three chosen centroid values into (5). The coefficient matrix has rows $(X, Y, 1)$; by (4) it tends to an invertible matrix. The right side, $-Y^2 - 4f\varepsilon X^2$, also has a finite limit. Hence $a_\varepsilon, b_\varepsilon, c_\varepsilon$ converge to finite real a, b, c. For any finite polygon, passage to the limit yields

$$Y^2 + aX + bY + c = 0, \qquad (X, Y) = C_k(P). \tag{6}$$

Divide by t^2 in the infinite-vertex configuration. Equations (1) and (2) give respectively

$$a = \frac{4f}{N} \quad (k = 0), \qquad a = \frac{4f}{3} \quad (k = 2).$$

In particular $a > 0$, so the limiting conic is a genuine parabola. Reflection in the x-axis preserves the fixed Poncelet family and commutes with each centroid. Subtracting (6) for (X, Y) and $(X, -Y)$ gives $2bY = 0$ throughout the locus. Since Y is unbounded, $b = 0$. This proves the stated equations for C_0, C_2.

These equations describe the entire finite parabolas, rather than only subarcs. Indeed, the space of unlabeled polygons is the quotient of the real outer conic by the order-N Poncelet step, hence is a circle. Only one orbit contains I. Removing it leaves an interval on which both centroids are continuous. At its two ends the single unbounded vertex has $t \to +\infty$ and $t \to -\infty$. By (1) and (2), Y has opposite infinite limits. The intermediate value theorem supplies every $Y \in \mathbb{R}$.

The perimeter centroid

For completeness, the identity used in [1, Remark 1] follows by decomposing the polygon into triangles from F. If its successive vertices are P_i and its edge lengths are ℓ_i, let $L = \sum_i \ell_i$. Each fan triangle has area $r\ell_i/2$, so

$$C_2 = \frac{\sum_i \ell_i (F + P_i + P_{i+1})}{3L}, \qquad C_1 = \frac{\sum_i \ell_i (P_i + P_{i+1})}{2L}.$$

These formulas also hold for a nested star traversal with winding multiplicity. Thus

$$C_1 = F + \frac{3}{2}(C_2 - F).$$

Substitution of the equation for C_2 gives the claimed equation for C_1, with $v_1 = 3v_2/2 + f/2$. This completes the proof.

Verification and scope

The accompanying verifier checks the projective conic identities, the dominant-triangle asymptotics, the centroid identity on exact asymmetric tangential polygons, and high-precision closed families in the $N = 3, 4, 5, 6$ cases. Numerical samples are checks of the formulas; the limit argument above proves the assertion for all N. The pentagon statement (MathDB 352453) is a special case of this result and receives no separate campaign credit. This proof does not use the parallel/perpendicular typo in the source's different polar conjecture.

Four proposed follow-up tasks

1. **Task 2.** Express v_0 and v_2 explicitly in terms of N, r/f, and standard elliptic-function invariants of the porism.

2. **Task 3.** For a circular caustic whose center is off the parabola's axis, determine the axes and vertices of all three centroid parabolas and characterize when their axes coincide.

3. **Task 4.** Determine, for every N, the vertex and area centroid loci of the polygons polar to this family with respect to $\mathcal{P}$, addressing Conjecture 5 of [1].

4. **Task 5.** Determine the leading constant and exponential rate in $1 - r_N/f$ as $N \to \infty$, where r_N is the circular caustic radius admitting the convex N-gon family.

References

[1] F. Bellio, R. Garcia, D. Reznik, *Parabola-Inscribed Poncelet Polygons Derived from the Bicentric Family*, KoG **26** (2022), 3–15, doi:10.31896/k.26.1; arXiv:2111.00979v3.

[2] R. Schwartz, S. Tabachnikov, *Centers of mass of Poncelet polygons, 200 years after*, Math. Intelligencer **38**(2) (2016), 29–34, doi:10.1007/s00283-016-9622-9; arXiv:1607.04766.

[3] A. C. Chavez-Caliz, *More about areas and centers of Poncelet polygons*, Arnold Math. J. **7** (2021), 91–106, doi:10.1007/s40598-020-00154-8; arXiv:2004.05404.